

Chương 5: VECTOR RIÊNG - TRỊ RIÊNG - CHÉO HÓA MA TRẬN

Vector riêng – Trị riêng

$$Au = \lambda u$$



$$\det(A - I\lambda) = 0$$

phương trình đặc trưng (tìm các trị riêng)



$$(A - I\lambda)u = 0$$

phương trình tìm các vector riêng



$$V_\lambda = \{u \in R^n \mid (A - I\lambda)u = 0\}$$

các không gian riêng ứng với các trị riêng

VD1: Tìm các trị riêng, vector riêng, và không gian riêng của $A = \begin{pmatrix} 3 & 2 \\ 3 & 8 \end{pmatrix}$

$$*\det(A - \lambda I_2) = 0 \rightarrow \det \begin{pmatrix} 3-\lambda & 2 \\ 3 & 8-\lambda \end{pmatrix} = 0$$

$$\rightarrow \lambda^2 - 11\lambda + 18 = 0 \rightarrow \begin{cases} \lambda = 2 \\ \lambda = 9 \end{cases}$$

$$*\lambda = 2 \rightarrow \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \rightarrow \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\rightarrow \begin{cases} x_1 = -2x_2 \\ x_2 \in R \end{cases} \rightarrow u_1 = (-2, 1) \text{ \& } V_\lambda = \{(-2, 1)\}$$

$$*\lambda = 9 \rightarrow \begin{pmatrix} -6 & 2 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \rightarrow \begin{pmatrix} -6 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\rightarrow \begin{cases} x_1 = x_2 / 3 \\ x_2 \in R \end{cases} \rightarrow u_2 = (1, 3) \text{ \& } V_\lambda = \{(1, 3)\}$$

Chéo hóa ma trận vuông $A_{n \times n}$

$$Au = \lambda u$$

Thu được các V_λ

Kiểm tra điều kiện chéo hóa
(tổng số các vector riêng = n)

Lập các ma trận $P = \begin{pmatrix} u_1 & u_2 & \dots & u_n \\ \downarrow & \downarrow & \dots & \downarrow \\ & & \dots & \end{pmatrix}$

Thu được ma trận chéo hóa

$$D = P^{-1}AP$$

VD2: Thực hiện chéo hóa ma trận

$$A = \begin{pmatrix} 2 & 3 \\ 0 & -1 \end{pmatrix} \rightarrow \det(A - \lambda I_2) = 0 \rightarrow \begin{cases} \lambda = -1 \\ \lambda = 2 \end{cases}$$

$$* \lambda_1 = -1 \rightarrow (A + I_2) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \rightarrow \begin{cases} x_1 = -x_2 \\ x_2 \in \mathbb{R} \end{cases}$$

$$\rightarrow u_1 = (-1, 1) \rightarrow V_{\lambda_1} = \{(-1, 1)\}$$

$$* \lambda_2 = 2 \rightarrow (A - 2I_2) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \rightarrow \begin{cases} x_1 \in \mathbb{R} \\ x_2 = 0 \end{cases}$$

$$\rightarrow u_2 = (1, 0) \rightarrow V_{\lambda_2} = \{(1, 0)\}$$

$$* \sum_{i=1}^2 \dim V_{\lambda_i} = 2 = n \rightarrow \text{chéo hóa được}$$

$$P = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$$



$$D = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$$

hoặc

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$



$$D = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$$

VD3: Tính A^{10} với

$$A = \begin{pmatrix} 2 & 3 \\ 0 & -1 \end{pmatrix}$$

$$* D = P^{-1}AP$$

$$\rightarrow D^n = (P^{-1}AP)^n = P^{-1}A^nP$$

$$\rightarrow A^n = PD^nP^{-1}$$

$$* P = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}, P^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

$$D^{10} = \begin{pmatrix} (-1)^{10} & 0 \\ 0 & 2^{10} \end{pmatrix} = \begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix}$$

$$\rightarrow A^{10} = PD^{10}P^{-1} = \begin{pmatrix} d_2 & -d_1 + d_2 \\ 0 & d_1 \end{pmatrix}$$

$$* P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, P^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$$D^{10} = \begin{pmatrix} 2^{10} & 0 \\ 0 & (-1)^{10} \end{pmatrix} = \begin{pmatrix} d_2 & 0 \\ 0 & d_1 \end{pmatrix}$$

$$\rightarrow A^{10} = PD^{10}P^{-1} = \begin{pmatrix} d_2 & -d_1 + d_2 \\ 0 & d_1 \end{pmatrix}$$

VD4: Chéo hóa & tính lũy thừa

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \rightarrow \det(A - \lambda I_3) = 0 \rightarrow \begin{cases} \lambda_1 = 1 \\ \lambda_2 = 4 \end{cases}$$

$$*\lambda_1 = 1 \rightarrow (A - I_3) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \rightarrow x_1 + x_2 + x_3 = 0$$

$$\rightarrow (-x_2 - x_3, x_2, x_3) \rightarrow x_2(-1, 1, 0) + x_3(-1, 0, 1)$$

$$\rightarrow \begin{cases} u_1 = (-1, 1, 0) \\ u_2 = (-1, 0, 1) \end{cases} \rightarrow V_{\lambda_1} = \{(-1, 1, 0), (-1, 0, 1)\}$$

$$*\lambda_2 = 4 \rightarrow (A - 4I_3) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \rightarrow x_1 = x_2 = x_3$$

$$\rightarrow (x_3, x_3, x_3) \rightarrow x_3(1, 1, 1) \rightarrow u_1 = (1, 1, 1)$$

$$\rightarrow V_{\lambda_2} = \{(1, 1, 1)\}$$

$$\sum \dim V_{\lambda} = n = 3 \rightarrow \text{chéo hóa được}$$

$$* P = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

↓

$$P^{-1} = \begin{pmatrix} -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \\ 1/3 & 1/3 & 1/3 \end{pmatrix}$$

↓

$$D = P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

$$* P = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \rightarrow P^{-1} = \begin{pmatrix} 1/3 & 1/3 & 1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{pmatrix}$$

$$\rightarrow D = P^{-1}AP = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$* P = \begin{pmatrix} -1 & 1 & -1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

↓

$$P^{-1} = \begin{pmatrix} -1/3 & 2/3 & -1/3 \\ 1/3 & 1/3 & 1/3 \\ -1/3 & -1/3 & 2/3 \end{pmatrix}$$

↓

$$D = P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A^{15} = ?$$

VD5: Tìm các trị riêng, vector riêng, và không gian riêng của

$$\begin{array}{llll} \text{a)} \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix} & \text{b)} \begin{pmatrix} 2 & 2 \\ 1 & 3 \end{pmatrix} & \text{c)} \begin{pmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{pmatrix} & \text{d)} \begin{pmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} \end{array}$$

VD6: Chéo hóa các ma trận sau (nếu được)

$$\begin{array}{lll} \text{a)} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{pmatrix} & \text{b)} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} & \text{c)} \begin{pmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} \end{array}$$

ỨNG DỤNG

(các VD ở đây được sưu tầm từ nhiều nguồn khác nhau)

VD7: Tính lũy thừa của ma trận

$$(a) \quad A = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} \Rightarrow ? : A^{10} \text{ \& } A^n$$

$$(b) \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 2 & -2 & 3 \end{bmatrix} \Rightarrow ? : A^{10} \text{ \& } A^n$$

VD8: Giải hệ phương trình vi phân tuyến tính bậc nhất

$$\mathbf{a)} \quad \begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} \rightarrow \begin{cases} \dot{x} = -2x \\ \dot{y} = -5y \\ x(0) = 3 \\ y(0) = 2 \end{cases} \rightarrow ? : x(t) \text{ \& } y(t) \rightarrow \begin{cases} x(t) = 3e^{-2t} \\ y(t) = 2e^{-5t} \end{cases}$$

$$\mathbf{b)} \quad \begin{Bmatrix} x(t) \\ y(t) \end{Bmatrix} \rightarrow \begin{cases} \dot{x} = 4x + 2y \\ \dot{y} = -x + y \\ x(0) = 1 \\ y(0) = 0 \end{cases} \rightarrow ? : x(t) \text{ \& } y(t) \rightarrow \begin{cases} x(t) = 2e^{3t} - e^{2t} \\ y(t) = -e^{3t} + e^{2t} \end{cases}$$

$$\mathbf{c)} \quad \begin{Bmatrix} T(t) \\ P(t) \\ C(t) \end{Bmatrix} \rightarrow \begin{cases} \dot{T} = -T + 4P + 3C \\ \dot{P} = -4T + 4P + C \\ \dot{C} = 5T + P - C \end{cases} \rightarrow ? : T(t), P(t), C(t)$$

VD9: Giải hệ phương trình vi phân tuyến tính bậc hai

$$\begin{aligned}m_1 \ddot{y}_1 &= -k_1 y_1 + k_2 (y_2 - y_1) \\m_2 \ddot{y}_2 &= -k_2 (y_2 - y_1)\end{aligned}$$

where dots denote derivatives with respect to time.

Write this system as a matrix equation $\ddot{Y} = AY$ and use the decoupling method to find Y if

(a) $m_1 = m_2 = 1, k_1 = 3, k_2 = 2$

and the initial conditions are $y_1(0) = 1, y_2(0) = 2, \dot{y}_1(0) = -2\sqrt{6}, \dot{y}_2(0) = \sqrt{6}$

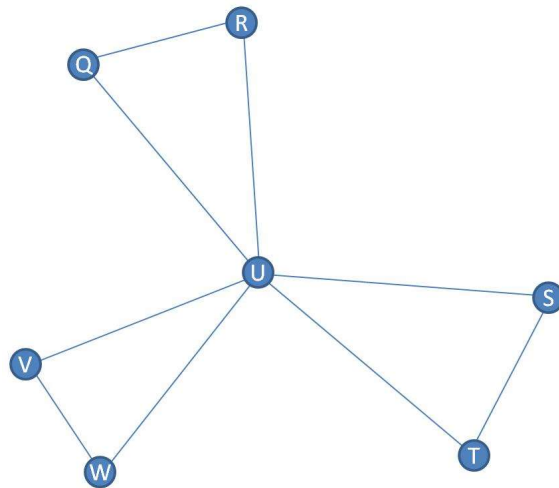
(b) $m_1 = m_2 = 1, k_1 = 6, k_2 = 4$

and the initial conditions are $y_1(0) = y_2(0) = 0, \dot{y}_1(0) = \sqrt{2}, \dot{y}_2(0) = 2\sqrt{2}$

$$\begin{aligned}\text{(a)} \quad Y &= \begin{bmatrix} \cos t - 2 \sin \sqrt{6}t \\ 2 \cos t + \sin \sqrt{6}t \end{bmatrix} & \text{(b)} \quad Y &= \begin{bmatrix} \sin \sqrt{2}t \\ 2 \sin \sqrt{2}t \end{bmatrix}\end{aligned}$$

VD10: Gould Index of Accessibility

a)



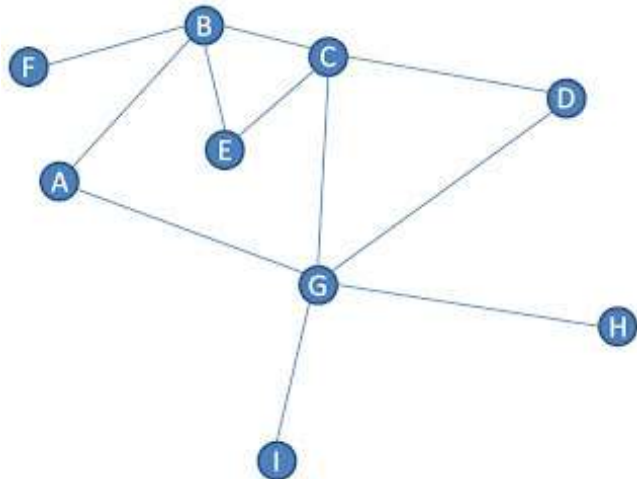
<i>Q</i>	<i>R</i>	<i>S</i>	<i>T</i>	<i>U</i>	<i>V</i>	<i>W</i>	
1	1	0	0	1	0	0	<i>Q</i>
1	1	0	0	1	0	0	<i>R</i>
0	0	1	1	1	0	0	<i>S</i>
0	0	1	1	1	0	0	<i>T</i>
1	1	1	1	1	1	1	<i>U</i>
0	0	0	0	1	1	1	<i>V</i>
0	0	0	0	1	1	1	<i>W</i>

The eigenvalues are: (2, 0, 4, -1, 0, 2, 0)

(0.3162, 0.3162, 0.3162, 0.3162, 0.6325, 0.3162, 0.3162)

(*Q, R, S, T, U, V, W*) = (0.125, 0.125, 0.125, 0.125, 0.25, 1.25, 1.25)

b)



The eigenvalues are:

(4.01, -1.37, 1.71, 1, -0.37, -0.56, 2.58, 1, 1)

(0.2941, 0.4097, 0.5023, 0.3249, 0.3026, 0.1359, 0.4770, 0.1583, 0.1583)

Normalizing we have (*A, B, C, D, E, F, G, H, I*) =

(0.1064, 0.1482, 0.1818, 0.1176, 0.1095, 0.0492, 0.1726, 0.0573, 0.0573)

VD11: POPULATION GROWTH

$$\left[0, \frac{L}{n}\right)$$

First age class

$$\left[\frac{L}{n}, \frac{2L}{n}\right)$$

Second age class

$$\vdots$$

$$\left[\frac{(n-1)L}{n}, L\right]$$

n th age class

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \begin{array}{l} \text{Number in first age class} \\ \text{Number in second age class} \\ \vdots \\ \text{Number in } n\text{th age class} \end{array}$$

Over a period of L/n years, the *probability* that a member of the i th age class will survive to become a member of the $(i+1)$ th age class is given by p_i , where

$$0 \leq p_i \leq 1, \quad i = 1, 2, \dots, n-1.$$

The *average number* of offspring produced by a member of the i th age class is given by b_i , where

$$0 \leq b_i, \quad i = 1, 2, \dots, n.$$

These numbers can be written in matrix form, as follows.

$$A = \begin{bmatrix} b_1 & b_2 & b_3 & \cdots & b_{n-1} & b_n \\ p_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & p_2 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & p_{n-1} & 0 \end{bmatrix}$$

Multiplying this **age transition matrix** by the age distribution vector for a specific time period produces the age distribution vector for the next time period. That is,

$$A\mathbf{x}_i = \mathbf{x}_{i+1}.$$

- a)** A population of rabbits has the following characteristics.
- a.** Half of the rabbits survive their first year. Of those, half survive their second year. The maximum life span is 3 years.
 - b.** During the first year, the rabbits produce no offspring. The average number of offspring is 6 during the second year and 8 during the third year.
- The population now consists of 24 rabbits in the first age class, 24 in the second, and 20 in the third. How many rabbits will there be in each age class in 1 year?

$$\begin{bmatrix} 304 \\ 12 \\ 12 \end{bmatrix} \begin{array}{l} 0 \leq \text{age} < 1 \\ 1 \leq \text{age} < 2 \\ 2 \leq \text{age} \leq 3 \end{array}$$

- b)** Find a stable age distribution vector for the population in Example 1.

$$A\mathbf{x} = \lambda\mathbf{x}. \quad \longrightarrow \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 16t \\ 4t \\ t \end{bmatrix} = t \begin{bmatrix} 16 \\ 4 \\ 1 \end{bmatrix}$$

$$\mathbf{x}_1 = \begin{bmatrix} 32 \\ 8 \\ 2 \end{bmatrix} \begin{array}{l} 0 \leq \text{age} < 1 \\ 1 \leq \text{age} < 2 \\ 2 \leq \text{age} \leq 3 \end{array} \quad \mathbf{x}_2 = A\mathbf{x}_1 = \begin{bmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} 32 \\ 8 \\ 2 \end{bmatrix} = \begin{bmatrix} 64 \\ 16 \\ 4 \end{bmatrix} \begin{array}{l} 0 \leq \text{age} < 1 \\ 1 \leq \text{age} < 2 \\ 2 \leq \text{age} \leq 3 \end{array}$$

VD12: MARKOV CHAIN

If the probability of moving from i to j in one time step is $\Pr(j|i) = P_{i,j}$, the stochastic matrix P j -th column element, e.g.,

$$P = \begin{bmatrix} P_{1,1} & P_{1,2} & \dots & P_{1,j} & \dots & P_{1,\alpha} \\ P_{2,1} & P_{2,2} & \dots & P_{2,j} & \dots & P_{2,\alpha} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ P_{i,1} & P_{i,2} & \dots & P_{i,j} & \dots & P_{i,\alpha} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ P_{\alpha,1} & P_{\alpha,2} & \dots & P_{\alpha,j} & \dots & P_{\alpha,\alpha} \end{bmatrix}.$$

Since the total of transition probability from a state i to all other states must be 1,

$$\sum_{j=1}^{\alpha} P_{i,j} = 1;$$

thus this matrix is a right stochastic matrix.

Example

- Consider the following model of population movement between a city and the suburbs:
 - each year 5% of city dwellers move the suburbs and
 - 3% of suburbanites move to the city
- If in 2001 58.2% of the population lived in the city and 41.8% lived in the suburbs, what is the **population distribution 20 years later?**

$$\text{year 2001: } \begin{cases} M = \begin{pmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{pmatrix} \\ x_0 = \begin{pmatrix} 0.582 \\ 0.418 \end{pmatrix} \end{cases} \rightarrow \text{year } k \text{ later 2021: } x_k = M^k x_0$$

$$P = \begin{pmatrix} 0.514496 & -0.707107 \\ -0.85749 & 0.707107 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 \\ 0 & 0.92 \end{pmatrix}$$

$$\lim_{k \rightarrow \infty} M^k x = \begin{pmatrix} 0.375 & 0.375 \\ 0.625 & 0.625 \end{pmatrix} x = \begin{pmatrix} 0.375 \\ 0.625 \end{pmatrix}$$

VD13: CÁC MỨC NĂNG LƯỢNG VÀ ORBITALS

$$H\Psi = E\Psi$$

